Quantum computing with neutral atoms: assignement

1 Introduction

A telecommunication company contacts you with the following situation. They have a certain number of antennas distributed over a geographical area, and each antenna uses a certain radio frequency to communicate. If two antennas are geographically close to each other, then they will interfere with each other if they use the same frequency (see Fig. 1). On the other hand, it's not possible to just assign a different frequency to each antenna, because the number of available frequencies is limited. Therefore, the client wants to assign a minimum number of different frequencies to their antennas, while minimizing interferences at the same time.

(a) (b) (c)

Figure 1: Schematic example of interference between antennas. (a) Bad frequency assignment. Two antennas are assigned the same frequency (represented by the color green) and they lie within the interference radius of each other (represented as a dashed circle around them). (b) The same antennas are assigned different frequencies (green and orange), so they will not interfere with each other even if they lie within the respective interference radius. (c) Two antennas can be assigned the same frequency (green) if they don't lie within each other's interference radius.

2 Materials

The client has 8 antennas. The physical (x, y) coordinates of the antennas are as follows:

```
antenna 1 (0, 0)
antenna 2 (3, 5.2)
antenna 3 (6, 0)
antenna 4 (9, -5.2)
antenna 5 (9, 0)
antenna 6 (9, 5.2)
antenna 7 (9, 10.4)
antenna 8 (12, 0)
```

The numbers are expressed in kilometers. Two antennas will interfere with each other if they are less than 8.7 km apart.

3 Assignment

Step 1
Explain how to model the *antenna frequency problem* as a *graph coloring* problem on *unit-disk graphs*.

Step 2
Design an analog quantum algorithm for neutral atoms that solves the graph coloring problem. The algorithm *does NOT* need to be optimal. You are free to choose the algorithm (adiabatic evolution, QAOA, others).

Step 3
Implement the algorithm in Pulser to find a solution to the client's problem.

Step 4 (optional)
Improve the implementation by including realistic error sources and hardware requirements in the algorithm.

Step 5
Write a short report detailing the work, clearly explaining the problem modeling, the algorithm, and justifying your choices.

4 Practicalities

The report can be written with the document editor of your choice, although Latex is preferred over other options like Word. The code should be properly commented, and it has to be provided as a repository on GitHub or GitLab, with a README explaining how it works. Clarity of exposition and readability of the code will be taken into account in the assessment. Optimality of the solution found by the algorithm is **NOT** necessarily the most important aspect. For example an algorithm that gives the correct solution for the given graph instance, but that is not based on correct and solid ideas, will be assessed negatively.